

Hongfei (Phil) Lin

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Summary

Current software development engineer at AWS Infrastructure Service. Experienced in full stack development and AWS technologies and using GenAI tools.

- **Programming Language**
Java, Python, SQL, Go, JavaScript, HTML, CSS Bootstrap, R, MATLAB
- **Software and Dev Tools**
Visual Studio Code, Cline, Kiro, Claude Code, IntelliJ, Jupyter Notebook

Education

University of Washington, Information School

Seattle, WA

Bachelor of Science in Informatics, Data Science

Expected: June 2022

Achievements: PCEP – Certified Python Programmer

Coursework: Data Structure and Algorithm, Database Design and Management, Data Visualization, Introduction to Machine Learning, Multivariable Calculus

Work Experience

AWS Infrastructure Service | *Software Development Engineer II* | Seattle, WA

April 2026 – Now

AWS Infrastructure Service | *Software Development Engineer* | Seattle, WA

August 2022 – April 2026

- Drove build planning enhancements like improving the interface with upstream systems, supporting S&OP processes and building new systems where legacy systems cannot support
- Worked with product managers and business customers to clarify requirements, design and recommend solutions and be independent in project executions solves difficult business domain problems and autonomously delivers results with high-quality production code
- Independently designed and implemented the full stack application that helped forecast the global power retirement plan for AWS data centers
- Raised the bar on operational excellence and drove continuous process improvements

Amazon Web Services | *Software Development Engineer Intern* | Seattle, WA

June 2021 – September 2021

- Designed and implemented a workflow for the team production runs
- Implemented and executed successfully AWS resources migration across eight team AWS accounts
- Configured two AWS Lambda functions to serve a specific purpose after resources migration
- Familiarized in using more than six AWS tech resources (Batch, CloudFormation, CloudWatch, etc..)
- Achieved 100% correctness in running results between the previous and the new workflow Beta runs

Project Experience

Simple API Web Server | *Backend Developer* | Seattle, WA

March 2021 – June 2021

- Built a web service using Go that summarizes the metadata in a web page with HTML tokenizing and JSON encoding techniques
- Implemented reusable, storage-agnostic packages for managing user accounts and session state using Redis and MySQL as the database
- Constructed RESTful APIs for sign-up, sign-in, sign-out, and user profile get/patch to the web server
- Dockerize the web client and API server and deployed the Docker container to the cloud via AWS EC2

Puzzle Games Formulation and Solving Techniques | *Backend Developer* | Seattle, WA

January 2021

- Formulated the “Farmer, Fox, Chicken and Grain” game as a state-space by creating a State class and an Operator class in Python
- Adapted Breadth-First Search, Depth-First Search algorithms that can trace back the states to find the ideal path for solving the game
- Implemented A-star algorithm to solve The Eight Puzzle game, and defined heuristic functions (using Hamming distance and Manhattan distance) to reduce the number of expanded states

Leadership Experience

Husky Help Organization | *Officer of Public Relations* | Seattle, WA

September 2018 - Present

- Led a team of 3 to raise funds from local restaurants to support activities, raised over \$300 profits
- Convinced restaurants to a mutually beneficial relationship – help us raise funds and increase the restaurant traffic
- Organized team meetings to brainstormed effective negotiation techniques of fund raising